

PROPOSED

Governance Proposal

PROPOSAL: Hangar AI Constitution — Complete Adoption Guide

Changeset ID: `build-adoption-guide-2026-001` **SDD Phase:** PROPOSE **Authority:** **ENG-11.1** (NON-NEGOTIABLE) **Status:** REMEDIATED — AWAITING ROUND 3 JURY DELIBERATION **Version:** 2.1 — Round 2 remediation (2026-05-01) **Round 1 Summary:** 79 objections across 6 jurors; 1 active BLOCK (**ENG-13.1**); 10 missing NN citations; 4 fabricated law titles; 3 misapplications; structural gaps in MVP scoping, personas, and compliance. All items remediated in v2.0. **Round 2 Summary:** 5 OBJECT / 1 APPROVE (Alexandra Pierce). 17 findings across 5 jurors. Remediated in this v2.1: Teaching Loop taxonomy corrected to AGENT.md §1.4 canonical; P3 learning outcome corrected; ENG-2.3 re-added (vertical slice applies to phased MVP delivery); **ENG-6.7** audit architecture substantively specified; **PRD-5.1** Timeline field added; PRD-5.3 experiment design completed (kill-if, control condition, guardrail metric); PRD-3.3 user story task added to Stage D; **BUS-3.6** explicitly scoped out; BUS-3.3 retention periods given calendar minimums; **BUS-9.3** DPO timing corrected to Stage B; Phase 0 gates corrected; CI/CD freshness workflow specified; JS test framework named; sonarqube-delta schema defined; evidence strength rubric defined.

1. Problem Statement (**PRD-1.2**)

Engineers, technical coaches, and senior architects adopting the Hangar AI Constitution encounter a fragmented onboarding experience. The laws, skills, avatars, and RAG model exist as isolated artifacts across `laws/`, `agent-skills/`, `avatars/`, and `docs/` — but there is no unified, navigable, persona-aware entry point connecting all components into a coherent adoption path.

Evidence Classification (PRD-1.5**):** WEAK — repository structural inspection only; no customer interviews on file at time of proposal. This is a known weak-evidence submission per **PRD-1.5**. A validation plan is provided below. Stage C (JTBD) WILL NOT begin until evidence is reclassified to Moderate (3+ structured interviews or equivalent quantitative proxy).

Observed Evidence (Repository Inspection):

- No landing page exists — `README.md` is the only structural index
- No persona-based quick start exists for distinct adopter types
- The RAG model visualization has no parent navigation or guide context
- Technical coaches cannot present a single URL that walks a new team through adoption
- The SDD workflow guide (PROPOSE→IMPLEMENT→ARCHIVE) has no interactive reference format
- The 168 laws are navigable only by reading raw markdown files
- Avatar selection requires manual exploration of `avatars/` — no wizard or decision path

PRD-1.2 Problem Template:

- **Who:** Technical coaches and senior architects onboarding new engineering teams to the Hangar AI Constitution
- **Problem:** No unified, navigable, persona-aware entry point; adopters must manually explore 10+ fragmented entry points
- **Evidence:** Repository inspection (WEAK — see validation plan below)
- **Frequency:** Every new team onboarding; estimated 3–5 engagements per quarter per coach
- **Severity:** HIGH — adoption friction compounds across every new team and project engagement

Validation Plan (**PRD-1.5** Weak Exception):

- Stage B task: Conduct 3+ structured interviews with technical coaches who have onboarded teams to the constitution
 - Stage B task: Collect quantitative proxy evidence from existing adoption logs, support channels, or ticket trackers
 - Evidence artifact: `hangar-ai-specs/changes/build-adoption-guide-2026-001/stage-b-evidence.md`
 - Gate: Stage C WILL NOT begin until `stage-b-evidence.md` is filed AND jury-validated at Moderate or Strong
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2. MVP Definition (PRD-5.1 🚫)

"MVPs SHALL be the smallest experiment to validate learning, not a crappy first version." — PRD-5.1

Riskiest Assumption: A unified landing page with persona-aware quick start paths (Technical Coach + Senior Architect) will meaningfully reduce time-to-first-successful-task compared to the current README-only entry point.

MVP Hypothesis: If we ship P1 (Landing Page) + P3 (Quick Start — Technical Coach and Senior Architect paths only), then first-successful-task time will decrease from estimated 45+ minutes (current baseline, to be measured in Stage B interviews) to < 20 minutes, measured during 3+ facilitator-observed walkthroughs with real adopters.

MVP Scope (NOW horizon only):

- P1: Landing Page — navigation hub, 3-persona entry points
- P3: Quick Start Guide — Technical Coach and Senior Architect paths only
- P2: Registered (already built — connect landing navigation)

Out of MVP Scope (NEXT/LATER — pending MVP validation):

- P4–P10 (deferred pending MVP validation signal)
- Engineer persona quick start path (deferred to NEXT)
- All analytics / behavioral telemetry (deferred behind BUS-4.5 PIA — LATER only)

Minimum Validation Threshold: MVP is considered validated when 2 of 3 observed walkthroughs achieve first-successful-task in < 20 minutes without facilitator prompting.

Timeline (PRD-5.1 🚫 — sixth required field): The MVP experiment runs for 4 weeks after P1 + P3 ship. Walkthroughs are conducted within the first 2 weeks of ship. Retention signal is measured at Week 4 (2 weeks post-walkthrough). The go/no-go call is made at Week 4 — no later. If walkthroughs cannot be scheduled within the first 2 weeks, the experiment clock is reset and the team files a risk register update before continuing.

Experiment Design (PRD-5.3):

FIELD	VALUE
Hypothesis	If we ship P1 + P3 (Technical Coach and Senior Architect paths), then first-successful-task time will decrease from the current README baseline (to be measured in Stage B) to < 20 min, for ≥ 2 of 3 real adopters
Experiment type	Concierge — facilitator-observed walkthroughs with real Technical Coaches and Senior Architects using the live guide
Control condition	Current README.md -only entry point — baseline task time formally measured in Stage B as part of assumption mapping; used as the comparison denominator in Stage E
Primary metric	First-successful-task time (stopwatch: landing page load → adopter declares task complete, without facilitator prompt)
Secondary metric	Adopter retention signal: returns to guide ≥ 2 times within 2 weeks of walkthrough (self-reported)
Guardrail metric	Adopter does not misidentify a non-negotiable (🚫) law — verified by Technical Coach juror post-walkthrough verbal check. A walkthrough that passes on task time but produces NN law misconceptions is not counted as a success.
Sample	n ≥ 3 real adopters (Technical Coaches or Senior Architects); n=3 is below statistical significance threshold — this is explicitly acknowledged; the experiment tests for signal, not significance
Segments	Included: Technical Coaches and Senior Architects who have worked on at least one AA project under the constitution. Excluded: Engineers (path not in MVP scope), adopters with < 1 week familiarity with the constitution
Ship if	≥ 2 of 3 walkthroughs achieve task time < 20 min AND ≥ 2 of 3 adopters pass the guardrail metric
Iterate if	1 of 3 walkthroughs succeed on task time; or all pass task time but 1 fails guardrail — team narrows scope or revises P3 path structure, then re-runs 1 additional walkthrough
Kill if	0 of 3 walkthroughs achieve task time < 20 min; or 2 of 3 fail the guardrail metric; or walkthroughs cannot be conducted within 2 weeks of P1+P3 ship date — if kill trigger fires, team re-enters Stage D and re-validates IA design
Minimum detectable effect	Not statistically computable at n=3. Effect size must be directionally large (> 50% reduction in task time) to be meaningful at this sample size. A reduction from 45 min to 44 min is not a signal.

Build-Measure-Learn Trigger (PRD-5.2): If kill criteria are met, the team pivots scope (narrow persona paths or restructure information architecture) before proceeding to P4–P10. Pivot path: re-enter Stage D; file `stage-d-evidence.md` v2; jury deliberation before new build.

Pivot Criteria (PRD-6.3): The experiment is killed and scope is pivoted if: (a) kill-if criteria are met per table above; or (b) Stage B interviews reveal the problem as stated in §1 is not validated — in this case the team returns to Stage B and re-files `stage-b-evidence.md` with updated problem framing before proceeding.

3. Full Solution Vision (Stage D — Aspirational, Pending MVP Validation)

Build a **10-page self-contained HTML adoption guide** that serves as the authoritative, navigable front-door to the Hangar AI Constitution for technical coaches and senior architects onboarding new teams.

Personas (PRD-3.1 — Minimum 5 Interviews Required Before Stage C)

Three human personas are defined. All three require evidence-based validation (5+ interviews per persona per PRD-3.1) before information architecture is confirmed in Stage C.

PERSONA	PRIMARY NEED	ENTRY POINT
Technical Coach	Facilitate team onboarding; present a single URL; apply Socratic guidance (AGENT.md §1.2–1.3) to lead engineers to discovery; use Discovery and Planning operating modes (AGENT.md §4.1, §4.2) as primary onboarding modes	P1 → P3 (Coach path) → P7, P8
Senior Architect	Evaluate architectural alignment; understand law precedence; map DDD patterns to constitution	P1 → P3 (Architect path) → P2, P4, P10
Engineer	Implement constitutionally; quick start on TDD gates, skill triggers, SDD workflow	P1 → P3 (Engineer path) → P5, P7, P8, P9

AI Agent is NOT a learning persona. An AI agent is an operating mode defined in AGENT.md §4 — it is a system component in the RAG model, not a human adopter navigating a guide. (Round 1 finding: Maya Chen OBJECT-3.)

New Project is NOT a persona. It is a context or trigger, not a person with goals and jobs-to-be-done. (Round 1 finding: Maya Chen OBJECT-4.)

ENG-13.1 Architecture Decision

All pages are self-contained HTML with no external stylesheet or script dependencies. Shared navigation, design tokens, and the AA gradient bar are inlined per `aa-artifact-render` production. No shared CSS files. This is a constitutional requirement, not an implementation preference.

ENG-13.2 Requirement — Citation Transparency on All Pages

Every law citation across all 10 pages SHALL use the `<span class="law-cite">` tooltip pattern, sourcing law title, NN status, and summary from `laws/index.yaml` YAML frontmatter. The `aa-artifact-render` tool handles this automatically when invoked with the `--laws-dir` flag.

JavaScript Test Infrastructure (ENG-4.1 🚫, Jordan OBJECT-16)

- Pages P4 (search JS) and P6 (wizard JS) contain inline JavaScript that must be tested per **ENG-4.1 🚫** Atomic TDD before ship.
- Test framework:** `Vitest` — chosen for zero-config setup, ES module support, and compatibility with inline-script extraction from HTML files
 - Runner invocation:** `npm vitest run` — runs once; exits with non-zero code on failure; suitable for CI
 - Test-to-inline-HTML pipeline:** Test files live alongside source (`p4-search.test.js` , `p6-wizard.test.js`). The build step extracts the inline `<script>` content from each HTML file into a separate module-compatible `.js` file, runs tests against that module, then re-inlines the verified script into the final HTML. The extracted `.js` files are build artifacts — not committed to the repository.
 - Gate G4 enforcement:** CI runs `npm vitest run` before the HTML is finalized. A failing test blocks the render. Setup task: added to Phase 3 in tasks.md (T3.3).

Generation Pipeline (ENG-11.3 — Spec Freshness)

Component	Source	Max Staleness	Trigger
P4 Laws Reference	laws/index.yaml + all laws/**/*.md frontmatter	30 days or on any law registry change	GitHub Actions on push to laws/
P5 Skills Catalog	agent-skills/*/index.yaml + AVATAR-RAG-INDEX.yaml	30 days or on skill index change	GitHub Actions on push to agent-skills/
P2 AI Model Viz	laws/index.yaml , AGENT.md , all skill index.yaml , AVATAR-RAG-INDEX.yaml	30 days or on any RAG source change	GitHub Actions on push to any source
P6 Avatar Wizard	AVATAR-RAG-INDEX.yaml + avatars/	30 days or on avatar index change	GitHub Actions on push to avatars/
P1, P3, P7-P10	Manual source — law IDs verified against registry	30 days or on registry version bump	Manual with freshness check gate

CI/CD enforcement: A GitHub Actions workflow validates source freshness on every PR. PRs that would advance a page beyond its staleness threshold are blocked until re-generated.

CI/CD Freshness Workflow Specification (ENG-11.3, Jordan OBJECT-15):

- **Workflow filename:** .github/workflows/freshness-check.yml
- **Trigger:** Every PR and every push to main
- **Staleness detection mechanism:** Each generated page (P2, P4, P5, P6) embeds a data-source-commit HTML attribute containing the git SHA of the source file at generation time. The workflow reads this attribute and compares it to the current HEAD SHA of the source. If the source has changed AND the page's embedded timestamp is > 30 days old, the check fails.
- **PR-blocking spec:** The workflow exits with a non-zero status code on staleness failure. The repository's branch protection rule on main includes freshness-check as a required status check — PRs cannot be merged with a failing freshness check.
- **Manual page coverage (P1, P3, P7-P10):** These pages have no auto-generated source. The workflow checks whether laws/index.yaml version has bumped since the page was last rendered (by reading a data-registry-version attribute). If the registry version changed, the workflow posts a PR comment flagging the page for manual review and fails the check until a re-render is committed.
- **CI/CD setup task:** Added to Phase 3 in tasks.md (T3.0 — before SonarQube provisioning).

Learning Arc (ENG-1.2 — Teaching Loop, AGENT.md §1.4)

The full guide follows the constitutional Teaching Feedback Loop as defined in AGENT.md §1.4. The six canonical stages are applied to the guide's page sequence:

Stage (AGENT.MD §1.4)	Pages	Teaching Loop Application
Observe — Identify what the adopter is trying to accomplish	P1	Adopter sees the full system at a glance; coach identifies adopter's entry goal
Guide — Ask questions to lead to discovery	P3	Adopter selects their persona path; Socratic coaching questions (per §1.3) replace prescriptive instructions
Explain — Provide constitutional context and rationale	P2, P4, P5	RAG model authority hierarchy, 168 laws with rationale, skills and their law bindings explained
Demonstrate — Show good vs. bad examples	P6, P7	Avatar wizard shows correct vs. incorrect selection; SDD workflow annotates compliant vs. non-compliant folder structures
Verify — Confirm understanding before proceeding	P8, P9	Adopter self-certifies compliance checklist; HARD_BLOCK conditions require adopter to articulate the correct response
Reinforce — Connect to broader principles for lasting learning	P10	Amendment process connects current law to constitutional evolution; adopter sees how their daily work feeds the governance system

Note on crosswalk: This application maps the Teaching Loop to a documentation guide context. The canonical loop is designed for agent-engineer pair interactions (AGENT.md §1.4). Pages P3, P8, and P9 are the most direct equivalents — the coach actively applies Guide, Verify, and Reinforce steps in facilitated walkthroughs.

Socratic Design Requirement: Every page SHALL include a "Questions to Explore" block with 3–5 open-ended coaching prompts. These prompts SHALL have no predefined answers within the page — they are discussion starters for a Technical Coach to use with their team.

Deliverables — 10-Page Guide

PAGE	TITLE	PURPOSE	TEACHING LOOP STAGE	PRIMARY LAWS	PER-PAGE LEARNING OUTCOME
P1	Landing Page	Navigation hub, constitution overview, 3-persona entry points	Observe	ENG-13.1  , PRD-1.1, ENG-13.2	Adopter identifies their persona and navigates to their entry point
P2	Constitutional AI Model Viz	RAG model, authority hierarchy, skill system, feedback loop (<i>BUILT</i>)	Explain	ENG-13.1  , ENG-13.2	Adopter describes all 5 RAG components and their authority relationships
P3	Quick Start Guide	Persona-based entry: Technical Coach / Senior Architect / Engineer	Guide	ENG-1.2, ENG-11.1  , ENG-13.2	Adopter identifies the first constitutional principle relevant to their context and can explain WHY it applies (not just WHAT it says)
P4	Laws Reference	Searchable 168 laws, 3 domains, NN flags, article structure	Explain	ENG-10.1  , ENG-4.1  , (search JS), ENG-13.2	Adopter locates any law by ID, title, or keyword and interprets NN status
P5	Skills Catalog	29+ skills, trigger phrases, law bindings, operating modes	Explain	ENG-11.1  , ENG-13.2	Adopter identifies 3+ skills relevant to their project context and finds their triggers
P6	Avatar Selection Wizard	Interactive: tech stack → laws + skills	Demonstrate	ENG-4.1  , (wizard JS), ENG-13.2	Adopter selects appropriate avatar and justifies selection against RAG model
P7	SDD Workflow Guide	PROPOSE→IMPLEMENT→ARCHIVE folder contract, ENG-11.2 checklist	Demonstrate	ENG-11.1  , ENG-11.2, ENG-13.2	Adopter scaffolds a compliant PROPOSE artifact with zero external help
P8	Compliance Checklist	Phase gates, SonarQube protocol, ENG-12.1 feedback loop	Verify	ENG-12.1  , ENG-12.2, ENG-12.3, ENG-13.2	Adopter completes a full phase gate without facilitator prompting
P9	Agentic Feedback Loop Guide	Human-in-the-loop contract, HARD_BLOCK conditions, gate diagrams	Verify	ENG-12.1  , ENG-13.2	Adopter recognizes a HARD_BLOCK condition and executes the correct response
P10	Amendment Process	ENG-10.x governance, law precedence, NN amendment path	Reinforce	ENG-10.1  , ENG-10.2, ENG-10.3, ENG-10.4, ENG-10.5, ENG-13.2	Adopter describes the amendment process and which laws require executive approval

4. Success Criteria (PRD-1.3 — Outcome-Driven)

Process Gates — All Required Before Phase Advances

GATE	REQUIREMENT	LAW AUTHORITY
G1	<code>PROPOSAL.md</code> and <code>tasks.md</code> rendered as HTML+PDF via <code>aa-artifact-render</code> before any jury deliberation	ENG-13.1  , ENG-13.3
G2	SonarQube provisioned, <code>.sonar-token</code> in <code>.gitignore</code> , dashboard open before Phase 1 IMPLEMENT begins	ENG-12.1 

GATE	REQUIREMENT	LAW AUTHORITY
G3	<code>stage-b-evidence.md</code> filed and jury-validated at Moderate evidence before Stage C begins	PRD-2.5 🚫, PRD-1.5 🚫
G4	Atomic TDD cycles confirmed for all JavaScript in P4 (search) and P6 (wizard) before ship	ENG-4.1 🚫
G5	Threat model filed in <code>compliance/threat-model.md</code> before Stage C	ENG-6.1 🚫
G6	Data classification table filed in <code>compliance/data-classification.md</code> before Stage C	BUS-3.1 🚫, ENG-6.4 🚫
G7	Risk assessment filed in <code>compliance/risk-register.md</code> before Stage C	BUS-6.1 🚫
G8	WCAG 2.1 AA accessibility audit + ADA / Section 508 / 14 CFR Part 382 determination obtained before final release	BUS-1.1 🚫, BUS-2.3 🚫
G9	Privacy Impact Assessment (PIA) completed and filed before any analytics feature ships	BUS-4.5, BUS-4.3 🚫
G10	SonarQube gate PASS + <code>hangar-ai-specs/evidence/sonarqube-delta.md</code> filed before archive	ENG-12.1 🚫
G11	<code>sonarqube-delta.md</code> filed at end of every session in <code>hangar-ai-specs/evidence/</code>	ENG-12.1 🚫

Evidence Strength Rubric (PRD-1.5 🚫, PRD-2.5 🚫)

The following rubric governs evidence classification at each stage gate. The G3 gate (Stage B exit) requires Moderate or Strong before Stage C begins. This rubric is non-negotiable — a juror may challenge any classification that does not meet the criteria.

CLASSIFICATION	CRITERIA	EXAMPLE
Weak	Repository / structural inspection only; no direct contact with real adopters; no interview on file; or: fewer than 3 interviews conducted; or: interviews conducted but findings not corroborated across participants	Stage A: README gap analysis (current stage)
Moderate	3–5 structured interviews conducted with real adopters from the target persona group; findings documented and internally consistent across ≥ 3 participants; OR a compelling quantitative proxy corroborating the qualitative gap (e.g., ≥ 20 support tickets with identical pattern); jury-validated by Priya Kapoor at Stage B gate	Stage B target: 3+ Technical Coach interviews with consistent "no single URL" finding
Strong	5+ structured interviews per persona; consistent themes confirmed across multiple data sources (qualitative + quantitative); no material contradictions across interview participants; quantitative evidence directionally agrees with qualitative findings	Stage C target: 5+ interviews per persona + ticket/log proxy + consistent JTBD framing

Evidence classification decisions are made by Priya Kapoor (Product Owner) and ratified by the full jury at each stage gate. The agent cannot self-classify above Weak without at least one real adopter interview on file.

SonarQube Delta Schema (ENG-12.1 🚫, Gates G10/G11)

The `hangar-ai-specs/evidence/sonarqube-delta.md` file MUST conform to the following schema at every session-end filing. The schema is defined in `compliance/sonarqube-delta-schema.md`. Required fields:

FIELD	TYPE	DESCRIPTION
<code>scan-id</code>	UUID	Unique identifier for the SonarQube scan run
<code>timestamp</code>	ISO 8601	Date and time of scan execution
<code>gate-result</code>	PASS or FAIL	SonarQube quality gate result
<code>delta-summary</code>	Dict	Changed metrics vs. previous scan (coverage %, new issues, resolved issues, code smells added/removed)
<code>human-reviewer</code>	String	Name of the person who opened the SonarQube dashboard and reviewed results
<code>sign-off</code>	Boolean + ISO 8601 timestamp	Human reviewer confirmation that results were reviewed before proceeding

FIELD	TYPE	DESCRIPTION
<code>session-context</code>	String	Which phase/task was in progress at time of scan

Filing a `sonarqube-delta.md` with missing required fields is equivalent to not filing it — Gate G11 is not satisfied until all fields are present.

Outcome Metrics

METRIC	TARGET	MEASUREMENT METHOD
First-successful-task time — adopter independently scaffolds a constitutionally valid task without facilitator prompt	< 20 min from landing page	3+ facilitator-observed walkthroughs; stopwatch start = landing page load, stop = adopter declares task complete
Per-page learning outcome met (adopter articulates outcome statement unprompted)	≥ 8 of 10 pages pass	Post-walkthrough verbal assessment by Technical Coach juror
ENG-13.1 compliance — all 10 pages self-contained HTML (no external dependencies)	10 / 10	Alexandra Pierce code review before release
Law citation accuracy — zero invented laws across all 10 pages	100%	Verified against <code>laws/index.yaml</code> v2.0.0 by Alexandra Pierce
ENG-13.2 tooltip coverage — all citations have inline tooltips with correct law metadata	100%	Automated count: <code>span.law-cite</code> count ≥ citation ID count in source
All 168 laws present in P4 Laws Reference	100%	Code-generated diff against <code>laws/index.yaml</code> ; zero manual entries
Adopter retention signal — adopter returns to guide for reference ≥ 2 times within 2 weeks of MVP walkthrough	≥ 2 of 3 MVP participants	Self-reported follow-up at Week 2 (no tracking pixels or analytics in MVP)
Jury unanimous approval	Required — ALL 6 jurors	See §6

5. ENG-10.1 Compliance Instrumentation (Constitution Metrics Collection)

- The adoption guide is a governed system artifact. Metrics collection instrumented per ENG-10.1 requirements:
- 1. **Compliance rate per law / team / project** — tracked via structured walkthrough observation forms (not behavioral analytics, avoiding PIA dependency in MVP)
 - 2. **Naming conventions** — all metric event names SHALL use the `aa.adoption_guide.<page>.<event>` pattern, consistent with org-wide telemetry conventions
 - 3. **Real-time collection** — metrics collected at the time of enforcement (walkthrough observation), not batched retroactively
 - 4. **PII exclusion** — no PII in metric dimensions; session IDs are ephemeral and non-identifiable
 - 5. **Production analytics** (LATER horizon only) — gated behind BUS-4.5 PIA; page-level events (load, navigation) classified as Internal per BUS-3.1; full analytics spec requires BUS-4.5 PIA before instrumentation

6. Jury — Deliberation Panel

All tasks require **UNANIMOUS approval** across all 6 jury members before progressing. **Alexandra Pierce holds BLOCKING VETO authority** on any non-negotiable law violation.

#	JUROR	ROLE	CONSTITUTION FOCUS	VETO
1	Tomás Reyes	Senior Architect	ENG-2.x: DDD, layered architecture, vertical slice integrity	—
2	Maya Chen	Technical Coach	ENG-1.2: Teaching-first, skill alignment, pedagogical clarity	—
3	Jordan Ellis	Staff Engineer / DevX	Developer experience, quick-start usability, adoption friction	—
4	Priya Kapoor	Product Owner	PRD-1.2, PRD-1.5, PRD-1.3: Outcome-Driven	—

#	JUROR	ROLE	CONSTITUTION FOCUS	VETO
5	Carlos Mendez	Compliance Officer	BUS-1.1 🚫, BUS-7.1 🚫, aviation regulations, data governance	—
6	Alexandra Pierce	Constitutional Lawyer (Sub-Agent)	ALL domains — NON-NEGOTIABLE watchdog. Active session monitor.	🚫 VETO

Deliberation Protocol

- Agent presents completed deliverable to jury; artifacts pre-rendered per ENG-13.1 🚫 and ENG-13.3 before jury sees them
- Each juror reviews cold against their constitutional domain — no prior context, no checklists pre-loaded
- Alexandra Pierce monitors ALL tasks continuously for NN violations; active at all times
- APPROVE** — Deliverable meets all relevant laws → unanimous across all 6 → proceed
- OBJECT** — Fixable deficiency found → agent remediates → re-deliberates
- BLOCK (Alexandra Pierce only)** — NON-NEGOTIABLE violation detected → agent **MUST** stop, cite violated law, remediate fully, re-present to full jury
- Minimum 2 deliberation rounds required — unanimous first-round APPROVE is constitutionally invalid
- Deliberation log maintained per BUS-7.1 🚫

Deliberation Log Specification (BUS-7.1 🚫)

Each log entry MUST capture all 6 required fields:

FIELD	CONTENT
Who	Juror name and role
What	Verdict (APPROVE / OBJECT / BLOCK) and action taken
When	ISO 8601 timestamp
Where	Proposal ID + stage/task identifier
Why	Law citation(s) and reasoning
Outcome	Net result (proceed / remediating / blocked)

Immutability mechanism (BUS-7.1 🚫 — corrected): Git branch protection policy MUST be configured with:

- Force-push disabled on `main` / `trunk`
- Branch deletion disabled on protected branches
- Signed commits required or SHA-256 commit signing enforced via CI
- No self-merge (require 1+ reviewer per PR)
- 7-year archival SLA: deliberation records in `hangar-ai-specs/archive/` are append-only under branch protection

Evidence integrity (BUS-7.2): A SHA-256 hash manifest (`compliance/evidence-manifest.sha256`) SHALL be maintained for all stage evidence files. Updated at each stage gate transition.

7. Discovery Stage-Gate Status (PRD-2.5 🚫)

STAGE	NAME	STATUS	EXIT ARTIFACT
A	Problem Framing	✅ COMPLETE	<code>stage-a-evidence.md</code> + <code>stage-a-evidence.html</code> filed
B	Assumption Mapping	🛑 BLOCKED — stage-b-evidence.md does not yet exist	<code>stage-b-evidence.md</code> (required)
C	Jobs-To-Be-Done	🕒 PENDING Stage B exit	<code>stage-c-evidence.md</code>
D	Solution Exploration	🕒 PENDING Stage C exit	<code>stage-d-evidence.md</code>
E	Prototype / Validation	🕒 PENDING Stage D exit	<code>stage-e-evidence.md</code>

STAGE	NAME	STATUS	EXIT ARTIFACT
F	Go/No-Go Decision	PENDING all prior stages	<code>stage-f-evidence.md</code>

Stage B deliberation CANNOT begin before `stage-b-evidence.md` is filed. The sequencing gate is a hard constitutional requirement per **PRD-2.5** — jury deliberation on a stage is not a substitute for the evidence artifact.

8. Roadmap (PRD-4.1 / PRD-4.2 — Outcome-Based, Now / Next / Later)

NOW — MVP Validation *Problem hypothesis: fragmented entry points cause measurable adoption friction*

- Outcome: ≥ 2 of 3 coached adopters complete first-successful-task in < 20 minutes unprompted
- Retention signal: ≥ 2 of 3 MVP participants return to guide for reference within 2 weeks
- Work: Stage B (assumption mapping + interviews) $\rightarrow$ Stage C (JTBD) $\rightarrow$ Stage D (IA design) $\rightarrow$ P1 + P3 (Tech Coach + Architect paths) built
- Compliance gates: G2, G3, G4, G5, G6, G7 must all pass before P1/P3 IMPLEMENT begins

NEXT — Full Guide *Condition: MVP validation threshold met (adoption friction confirmed, retention signal positive)*

- Outcome: Technical Coach can deliver a complete team onboarding session using the guide as the sole resource, without any manual document lookups
- Work: P4 (Laws Reference), P5 (Skills Catalog), P6 (Avatar Wizard), P7 (SDD Workflow), P8 (Compliance Checklist), P9 (Agentic Feedback Loop), P10 (Amendment Process)
- Engineer persona quick start path added to P3
- Gate: MVP retention signal met; SonarQube gate PASS; G8 (accessibility) must pass before NEXT release

LATER — Continuous Evolution *Condition: Full guide in production; at least one full adoption cycle observed*

- Outcome: Guide stays current with constitution amendments without manual intervention; zero adopter-observed staleness incidents per quarter
- Work: CI-driven source freshness (ENG-11.3 auto-regeneration triggers); amendment process auto-update
- Analytics: page-level behavioral telemetry — ONLY after BUS-4.5 PIA is completed and filed; NOT before

9. Regulatory Scope Declaration (BUS-2.x)

This declaration is required before Stage C per **BUS-2.1** , **BUS-2.2** , **BUS-2.3** .

LAW	SCOPE DECISION	RATIONALE
BUS-2.1 FAA Compliance Law (FAR Part 25, FAR Part 117, DO-178C)	OUT OF SCOPE for guide content itself. The guide documents the Hangar AI Constitution governance system — it does not control avionics software or crew scheduling software directly. Projects using the guide that produce avionics-adjacent software remain bound by FAR Part 117 / DO-178C through their own SDD proposals.	The adoption guide is a documentation artifact, not a flight-critical or crew-scheduling software system.
BUS-2.2 Control Framework Law	IN SCOPE — Controls governing the adoption guide's own change process (jury approval, branch protection, evidence filing) are documented and mapped in <code>compliance/raci.md</code> and <code>compliance/risk-register.md</code> .	Every artifact produced under the constitution is subject to the control framework.
BUS-2.3 DOT Consumer Protection Law (refund obligations, accessibility, denied boarding)	ACCESSIBILITY IN SCOPE — 14 CFR Part 382 (nondiscrimination for air travelers with disabilities) applies to American Airlines customer-facing digital products. The adoption guide is an internal-only artifact; however, accessibility compliance (WCAG 2.1 AA) is required per BUS-1.1 Priority Hierarchy (Legal first). Gate G8 mandates an ADA / Section 508 / Part 382 determination before final release. Other DOT consumer protection provisions (refunds, fare transparency, denied boarding) are out of scope for a governance guide.	Internal artifact; fare/refund/boarding provisions inapplicable. Accessibility provision applies to all digital surfaces at AA per legal guidance.
BUS-2.4 Evidence Collection Law	IN SCOPE — Compliance evidence for this proposal (stage evidence files, deliberation log, SonarQube delta) is retained in <code>hangar-ai-specs/</code> per ENG-11.1 and archived per BUS-7.1 .	Standard evidence retention applies.

Full regulatory scope document: `compliance/regulatory-scope.md` (to be filed before Stage C, per G5/G6/G7)

10. Data Classification Plan (BUS-3.1 🚫)

All data assets produced or processed by this project are classified below per BUS-3.1 🚫 .

DATA ASSET	CLASSIFICATION	RATIONALE	RETENTION
PROPOSAL.md, tasks.md, PROGRESS.md	Internal	Internal governance documents; no PII; not for external publication	Minimum 3 years post-archive per BUS-2.4; deleted no earlier than 3 years after <code>hangar-ai-specs/archive/</code> move
Stage evidence files (stage-a through stage-f)	Internal	Interview notes may include attributed quotes (de-identified before filing)	Minimum 3 years post-archive per BUS-2.4
Jury deliberation log (SQL session DB)	Internal	Contains juror names (personas) and verdict reasoning; no real PII	Online: SDD lifecycle + 1 year; Archived: 7 years per BUS-7.1 🚫 minimum, append-only in <code>hangar-ai-specs/archive/</code> under branch protection
SonarQube delta evidence	Internal	Code quality metrics; no PII	7 years per BUS-7.1 🚫 evidence policy (governs all compliance evidence for this project)
walkthrough observation forms (MVP measurement)	Internal	De-identified; no names recorded; adopter is identified only by role	90 days post-MVP validation
Production analytics events (LATER horizon)	Internal / Confidential (TBD — requires PIA)	Behavioral page events; potential PII if user is logged in	TBD pending PIA
Compliance artifacts (threat model, risk register, RACI)	Confidential	Contains organizational risk posture information	Minimum 3 years post-archive per BUS-2.4; threat model and risk register: 7 years per BUS-7.3 audit readiness

Full data inventory: `compliance/data-classification.md` (to be filed before Stage C per Gate G6)

11. Compliance Ownership & RACI (BUS-1.3)

Every compliance obligation under this proposal has a named accountable owner per BUS-1.3.

OBLIGATION	ACCOUNTABLE OWNER	RESPONSIBLE	CONSULTED	INFORMED
Constitutional compliance (all NN laws)	Alexandra Pierce (sub-agent)	Agent	All 6 jurors	Constitution maintainer
Privacy / data subject rights (BUS-4.3 🚫 , BUS-4.5)	Carlos Mendez (Compliance Officer)	Agent	Alexandra Pierce	DPO
Accessibility gate (BUS-2.3 🚫 , G8)	Carlos Mendez	Agent	BUS-1.1 authority	AA Legal
Audit trail integrity (BUS-7.1 🚫)	Carlos Mendez	Agent	Alexandra Pierce	Repo admin
MVP outcome measurement (PRD-5.1 🚫 , PRD-6.2 🚫)	Priya Kapoor	Agent	Jordan Ellis	Tech Coach stakeholders
Evidence evidence quality (PRD-1.5 🚫)	Priya Kapoor	Agent	Maya Chen	Constitution maintainer
Pedagogical quality (ENG-1.2)	Maya Chen	Agent	Priya Kapoor	Tech Coach community
Architecture compliance (ENG-4.1 🚫 , ENG-6.1 🚫)	Tomás Reyes	Agent	Alexandra Pierce	Platform team
SonarQube gate (ENG-12.1 🚫)	Jordan Ellis	Agent	Tomás Reyes	CI/CD owner

Full RACI matrix: `compliance/raci.md` (to be filed before Stage C per Gate G7)

12. Risk Assessment Reference (BUS-6.1 🛑)

Project-specific risks identified at PROPOSE stage. Full assessment in `compliance/risk-register.md`.

RISK	LIKELIHOOD	IMPACT	CONTROL
Weak evidence prevents Stage C advancement; MVP scope under-validated	HIGH	HIGH	Stage B gate (G3) is hard-blocking; minimum 3 interviews before Stage C
P4 search JS or P6 wizard JS contains untested code paths	MEDIUM	HIGH	ENG-4.1 🛑 TDD gate (G4); mutation testing before ship
analytics feature shipped without PIA	MEDIUM	CRITICAL	G9 hard-blocks analytics; all analytics in LATER horizon only
Law registry staleness causes incorrect citations in guide	MEDIUM	HIGH	ENG-11.3 generation pipeline; CI freshness check on every PR
Jury deliberation log tampered or lost	LOW	HIGH	BUS-7.2 SHA-256 manifest + branch protection policy (BUS-7.1 🛑)
Accessibility defect discovered post-release	LOW	HIGH	G8 mandatory before final release; not in LATER
Constitution amendment during IMPLEMENT causes guide staleness	MEDIUM	MEDIUM	ENG-11.3 freshness trigger; 30-day max staleness window
Registry integrity anomaly (BUS-2.2 / BUS-3.1 / BUS-6.1 comment mismatch in laws/index.yaml)	CONFIRMED	MEDIUM	Escalated to constitution maintainer; tracked in risk register

Full risk register: `compliance/risk-register.md` (to be filed before Stage C per Gate G7)

13. Law Citations (ENG-11.2 — Required)

All citations verified against `laws/index.yaml` v2.0.0. Zero fabricated titles. 36 laws cited.

Engineering Constitution

LAW ID	TITLE	CLASS	ROLE IN THIS PROPOSAL
ENG-1.2	AI-Engineer Pairing Law	Standard	Guide embodies teaching-first — every page coaches, not just informs; learning arc required
ENG-2.3	Vertical Slice Architecture Law	Standard	The phased MVP delivery (P1+P3 as a fully validated, independently shippable slice before P4–P10) is the vertical slice pattern applied to a documentation project. Each slice (MVP slice: P1+P3; NEXT slice: P4–P7; LATER: P8–P10) delivers end-to-end adopter value without requiring subsequent slices to be useful. The MVP slice is tested and jury-approved before the next slice begins — this is the constitutional vertical slice principle. (Removed in v2.0 — re-added per Tomás Reyes Round 2 OBJECT-13: deletion without scoping rationale is not permitted.)
ENG-4.1	Atomic TDD Law	🛑 NN	P4 (search JS) and P6 (wizard JS) are software — TDD atomic cycles required; Gate G4
ENG-6.1	Security by Design Law	🛑 NN	Threat model required before Stage C; guide exposes amendment paths, HARD_BLOCK conditions, SonarQube gate protocol; Gate G5
ENG-6.4	Data Protection Law	🛑 NN	Analytics forward commitment requires data classification; all data assets classified §10; Gate G6
ENG-6.7	Audit Trail Law	🛑 NN	Audit architecture for this pipeline context: Logged operations: phase gate transitions (G1–G11), <code>aa-artifact-render</code> invocations, stage evidence file creation/update, jury deliberation entries, git commit signing events. Storage: git history (append-only under branch protection — force-push disabled, force-delete disabled per §6) + SQL session checkpoint archive (session store). Record structure per audit event: <code>{ timestamp: ISO 8601, operation: string, actor: juror-id or "agent" or "ci", artifact-id: changeset + filename, outcome: "PASS"/"FAIL"/"APPROVE"/"OBJECT"/"BLOCK", correlation-id: UUID linking related gate events }</code> . Immutability: branch protection prevents git history rewrite; SHA-256 manifest (<code>compliance/evidence-manifest.sha256</code>) verifies artifact integrity at each gate. Scope boundary: this is an internal pipeline artifact, not a flight-safety or financial system — immutability

LAW ID	TITLE	CLASS	ROLE IN THIS PROPOSAL
			is achieved through git branch protection + SHA-256 verification, not a dedicated tamper-evident log service. This is explicitly distinct from BUS-7.1 's deliberation log obligation (which governs jury records specifically).
ENG-10.1	Constitution Metrics Collection Law	🚫 NN	Adoption guide is a governed system artifact; compliance tracking instrumented per §5: naming conventions, real-time collection, no PII in dimensions
ENG-10.2	Enforcement Tracking Law	Standard	P10 Amendment Process sources enforcement tracking requirements from this law
ENG-10.3	Compliance Reporting Law	Standard	P10 Amendment Process sources compliance reporting cadence from this law
ENG-10.4	Constitution Health Dashboard Law	Standard	P10 references health dashboard requirements
ENG-10.5	Law Effectiveness Measurement Law	Standard	P10 references effectiveness measurement requirements
ENG-11.1	Hangar SDD Law	🚫 NN	This PROPOSAL.md IS the PROPOSE stage of ENG-11.1 ; <code>hangar-ai-specs/</code> folder contract defined §14
ENG-11.2	Proposal Completeness Law	Standard	This §13 satisfies the law citation requirement
ENG-11.3	Spec Freshness Law	Standard	Guide must reflect current constitution state; generation pipeline defined §3; max staleness 30 days
ENG-12.1	Agentic Feedback Loop Law	🚫 NN	SonarQube MUST be provisioned before Phase 1 IMPLEMENT begins; dashboard reviewed at every phase transition; session-end <code>sonarqube-delta.md</code> filed; Gate G2, G10, G11
ENG-12.2	Dashboard-First Development Law	Standard	SonarQube dashboard open throughout each session; not just at API call
ENG-12.3	External Referee Law	Standard	Agent cannot self-certify compliance; SonarQube provides objective verdict
ENG-13.1	Artifact Rendering Standard	🚫 NN	All 10 pages and this PROPOSAL.md rendered via <code>aa-artifact-render</code> before deliberation; <code>tasks.md</code> also rendered (active BLOCK resolved)
ENG-13.2	Citation Transparency Law	Standard	All law citations across all 10 pages use <code></code> tooltip pattern sourced from law YAML frontmatter
ENG-13.3	PDF Reproducibility Law	Standard	Gate reviews use <code>aa-artifact-render --pdf</code> ; browser-print PDFs not acceptable; Gate G1

Product Constitution

LAW ID	TITLE	CLASS	ROLE IN THIS PROPOSAL
PRD-1.1	Customer-Centric Law	Standard	P1 Landing Page anchored in adopter needs, not internal preferences
PRD-1.2	Problem-First Law	🚫 NN	§1 validates problem before solution; solution architecture deferred to Stage D
PRD-1.3	Outcome-Driven Law	Standard	Success criteria in §4 defined as outcomes, not feature counts

LAW ID	TITLE	CLASS	ROLE IN THIS PROPOSAL
PRD-1.4	Continuous Discovery Law	Standard	Discovery is never complete; Stage F Go/No-Go is not the end of learning
PRD-1.5	Evidence-Based Decision Law	🚫 NN	Weak evidence acknowledged in §1; validation plan filed; Stage C gated on Moderate evidence
PRD-2.1	Problem Validation Law	Standard	Problem must be jury-validated before IA design in Stage D
PRD-2.2	Assumption Mapping Law	Standard	Stage B maps all adopter assumptions; filed in <code>stage-b-evidence.md</code>
PRD-2.3	Jobs-to-be-Done Law	Standard	Stage C frames adopter needs as jobs for 3 personas
PRD-2.4	Competitive Analysis Law	Standard	Stage B includes alternatives analysis (existing README, other constitution guides)
PRD-2.5	Discovery Stage-Gate Law	🚫 NN	Stages A→F tracked in §7; Stage B hard-blocking gate (G3)
PRD-3.1	Persona Development Law	Standard	3 human personas; evidence-based (5+ interviews each) required before Stage C
PRD-3.2	Journey Mapping Law	Standard	Journey maps required before IA design in Stage D
PRD-3.3	User Story Law	Standard	Stories with acceptance criteria required in Stage D
PRD-3.4	Experience Principles Law	Standard	Guiding experience principles defined in Stage D
PRD-4.1	Outcome-Based Roadmap Law	Standard	§8 roadmap communicates outcomes per horizon, not feature lists
PRD-4.2	Now/Next/Later Framework Law	Standard	§8 uses Now/Next/Later time horizons, not fixed dates
PRD-4.3	Dependency Management Law	Standard	Stage gate dependencies explicitly documented in §7 and tasks.md
PRD-4.4	Roadmap Communication Law	Standard	§8 roadmap is appropriate for technical coaches + architects audience
PRD-5.1	MVP Law	🚫 NN	§2 defines MVP: riskiest assumption, smallest experiment, validation threshold
PRD-5.2	Build-Measure-Learn Law	Standard	BML trigger defined in §2: if MVP threshold not met, pivot before P4–P10
PRD-5.3	Experiment Design Law	Standard	Full experiment design in §2: hypothesis, experiment type (Concierge), control condition (README baseline), primary/secondary/guardrail metrics, sample, segments, ship-if, iterate-if, kill-if, minimum detectable effect
PRD-6.1	PMF Definition Law	Standard	Adoption-market fit to be measured post-MVP, not assumed
PRD-6.2	Retention Over Acquisition Law	🚫 NN	Retention signal defined in §4: ≥ 2 of 3 MVP participants return within 2 weeks
PRD-6.3	Pivot Criteria Law	Standard	Pivot criteria defined in §2: kill-if conditions trigger Stage D re-entry; stage B contradiction triggers problem revalidation before any build proceeds

Business Constitution

LAW ID	TITLE	CLASS	ROLE IN THIS PROPOSAL
BUS-1.1	Priority Hierarchy Law	🚫 NN	Legal > Safety > Privacy > Security > Business Continuity > Efficiency honored across all deliverables; accessibility pre-release (Gate G8), not deferred
BUS-1.2	Risk-Based Approach Law	Standard	Governance decisions proportionate to risk; risk-based triage in §12
BUS-1.3	Accountability Law	Standard	Named compliance owners defined in §11; RACI in <code>compliance/raci.md</code>
BUS-1.4	Transparency Law	Standard	Compliance gaps documented in §12 and <code>compliance/risk-register.md</code>
BUS-2.1	FAA Compliance Law	🚫 NN	Scoped OUT for guide content; scoped IN for projects built using the guide — see §9
BUS-2.2	Control Framework Law	🚫 NN	Controls documented in <code>compliance/raci.md</code> ; jury approval is a control; branch protection is a control
BUS-2.3	DOT Consumer Protection Law	🚫 NN	Accessibility (14 CFR Part 382) in scope; Gate G8 mandatory; other provisions out of scope for internal guide
BUS-2.4	Evidence Collection Law	Standard	All stage evidence retained in <code>hangar-ai-specs/</code> ; SHA-256 manifest in <code>compliance/evidence-manifest.sha256</code>
BUS-3.1	Data Classification Law	🚫 NN	All data assets classified in §10; full table in <code>compliance/data-classification.md</code>
BUS-3.2	Data Inventory Law	Standard	Data inventory in <code>compliance/data-classification.md</code> with owner, location, retention
BUS-3.3	Data Retention Law	Standard	Retention schedule with explicit calendar minimums per asset defined in §10 table; deliberation log: 7-year archive per BUS-7.1 🚫 ; SDD artifacts: 3-year minimum per BUS-2.4
BUS-3.6	Monetary Precision Law	🚫 NN	SCOPED OUT — NOT APPLICABLE. The adoption guide is a 10-page static HTML governance documentation artifact. It contains no monetary computations, no loyalty-currency values (AAdvantage miles, EQDs, points), no fare arithmetic, no revenue figures, and no financial or loyalty quantities of any kind across P1–P10. BUS-3.6's decimal precision and rounding requirements do not apply. Explicitly scoped out per compliance record requirements.
BUS-4.1	Privacy by Design Law	Standard	Analytics uses Privacy by Design 7 principles if/when shipped (LATER horizon)
BUS-4.2	Consent Management Law	Standard	Analytics consent mechanism required before production analytics ship
BUS-4.3	Data Subject Rights Law	🚫 NN	Behavioral analytics data (LATER) subject to access, rectification, erasure, portability rights; Gate G9
BUS-4.4	Privacy Notice Law	Standard	Notice required before any analytics feature ships
BUS-4.5	Privacy Impact Assessment Law	Standard	PIA required before analytics ships; Gate G9; LATER horizon only
BUS-6.1	Risk Assessment Law	🚫 NN	Project-specific risk assessment in §12; full register in <code>compliance/risk-register.md</code> ; Gate G7
BUS-6.2	Risk Register Law	Standard	Risk register maintained in <code>compliance/risk-register.md</code>
BUS-7.1	Audit Trail Law	🚫 NN	Deliberation log maintained per §6; branch protection policy required for immutability claim
BUS-7.2	Evidence Integrity Law	Standard	SHA-256 hash manifest in <code>compliance/evidence-manifest.sha256</code>
BUS-7.3	Audit Readiness Law	Standard	Documentation current; archive policy per ENG-11.1 🚫

LAW ID	TITLE	CLASS	ROLE IN THIS PROPOSAL
BUS-7.4	Internal Audit Law	Standard	Post-release audit scheduled; findings tracked in <code>PROGRESS.md</code>
BUS-9.1	Incident Classification Law	Standard	Incidents classified P1-P4 per response plan in <code>compliance/incident-response.md</code>
BUS-9.2	Incident Response Law	Standard	Defined response process in <code>compliance/incident-response.md</code>
BUS-9.3	Breach Notification Law	🚫 NN	72-hour GDPR reporting window documented; DPO contact chain in <code>compliance/incident-response.md</code>
BUS-9.4	Post-Incident Review Law	Standard	Post-mortems required for all significant incidents; findings feed risk register

14. SDD Folder Contract (ENG-11.1)

```

hangar-ai-specs/changes/build-adoption-guide-2026-001/
├── PROPOSAL.md           ↳ This document (PROPOSE stage artifact)
├── PROPOSAL.html         ↳ Rendered per ENG-13.1 🚫 (aa-artifact-render)
├── PROPOSAL.pdf          ↳ Rendered per ENG-13.3 (aa-artifact-render --pdf)
├── tasks.md              ↳ Atomic task breakdown with checkboxes
├── tasks.html            ↳ Rendered per ENG-13.1 🚫
├── tasks.pdf             ↳ Rendered per ENG-13.3
├── PROGRESS.md           ↳ Phase tracking and gate evidence
├── stage-a-evidence.md    ↳ ✅ Stage A exit artifact (PRD-2.5 🚫)
├── stage-a-evidence.html  ↳ ✅ Rendered
├── stage-b-evidence.md    ↳ 🚫 NOT YET FILED – Stage B gate blocks Stage C
├── compliance/           ↳ BUS-1.3, BUS-2.x, BUS-3.1, BUS-6.1, ENG-6.1 artifacts
│   ├── regulatory-scope.md ↳ BUS-2.x declaration (Gate G5 dependency)
│   ├── data-classification.md ↳ BUS-3.1 🚫 classification table (Gate G6)
│   ├── threat-model.md    ↳ ENG-6.1 🚫 threat model (Gate G5)
│   ├── risk-register.md   ↳ BUS-6.1 🚫 risk register (Gate G7)
│   ├── raci.md            ↳ BUS-1.3 accountability matrix
│   ├── incident-response.md ↳ BUS-9.3 🚫 incident response + DPO contact chain
│   └── evidence-manifest.sha256 ↳ BUS-7.2 SHA-256 hash manifest
└── hangar-ai-specs/evidence/
    └── sonarqube-delta.md  ↳ ENG-12.1 🚫 session-end gate evidence (Gate G10, G11)

```

Upon completion: move to `hangar-ai-specs/archive/YYYY-MM-DD-build-adoption-guide-2026-001/`

Proposal v2.1 — Remediated per Round 2 jury findings (2026-05-01) 40 law citations — all verified against `laws/index.yaml` v2.0.0 — zero fabricated titles Authored per **ENG-11.1** 🚫 / ENG-11.2 · Round 3 jury deliberation required · Alexandra Pierce monitoring active